

Introduction to Programming COMP101

Tutorial-05 Pseudocode Selection

Objectives:

Exercise-01 Pseudocode Selection to Python

Exercise-02 Pseudocode to Selection to Python

Introduction

Selection (if-then-else) is a construct that alters the flow of control of a program.

It does this by allowing some statements to execute and others not to execute.

Pseudocode is not language-specific so even though our platform (Python) does not use the 'then' statement but uses the 'elif' statement, it is the general approach to identifying the logic of selection across all programs that prevails.

Hence we refer to the language sub-set of our pseudocode to design the program, i.e we use if – then – else and we don't use elif as elif is specific to Python.

Exercise-01: Pseudocode Selection

A die (singular of the plural 'dice') is a six-sided cube with each side of the cube showing an integer 1 to 6. There is one unique integer on each side.

A user 'throws' two die. If they throw a 'double six' they are told they can play the game again. At this point the program ends.

If it not a 'double six' they are told to pass the dice to another player.
At this point the program ends.

#no initialisation

1. input

1.1 input integer1

1.2 input integer2

2. process

2.1 calc total = integer1 + integer2

2.2 if total = 12 then

2.2.1 print 'Throw again'

2.3 else

2.3.1 print 'pass dice'

3. output

3.1 print game over

Notes Exercise-01:

a) No initialisation included due to trivial design with limited i/o.
Depending on the logic and the specification, it may not need to include all sections of the design

b) Consider the logic in line 2.2 of the design.
Total = 12 can only be achieved when both die 'show' a six.
It is a more efficient approach than:
2.2 If integer1 = 6 and integer2 = 6 then
 2.2.1 print 'Throw again'

This another reason (other than clarifying the logic) to number the lines – we can go straight to the line in question to review it.

Ex-01 Python SOLUTION

#tutorial-04_ex-01 solution.py

```
die1 = int(input("Enter score for die1: "))  
die2 = int(input("Enter score for die2: "))
```

```
total = die1 + die2
```

```
if total == 12:  
    print("Throw again")  
else:  
    print("Pass the dice to the next player")
```

In the actual code you could consider using import random to simulate the throw of the die

You will come across variation on the design structure but all variations must identify the same logic and process required.

As there are no officially recognised standards, many programmer are asked to adopt the 'House Style' when they program for a software house.

Our house style is as in the tutorials, but an employer will have their own approach to design to enforce consistency between programming teams.

e.g. you may find something like:

1. input score
2. if score >=90
3. then
4. print A

5. if score >=80
6. then
- 7 print B

8. if score >=70
9. then
10. print C

11. if score >=60
12. then
13. print D

14. else
15. print Fail

Exercise-02: Pseudocode Selection

Alphabetic grades A, B, C, D and F are awarded to assessment work depending on the numeric score achieved.

Produce a Python program from the following design;
Use one if-statement, one else-statement and as many elif-statements as needed.

```
# no initialisation
1 input
    1.1 input score
2 process
    2.1 if score >= 90 then
        2.1.1 print ("A")
    2.2 if score >= 80 then
        2.2.1 print ("B")
    2.3 if score >=70 then
        2.3.1 print ("C")
    2.4 if score >= 60 then
        2.4.1 print ("D")
    2.5 else
        2.5.1 print ("F")
```

NOTES Exercise-02:

- ‘then’ is okay in design.
 - Remember Python does not use ‘then’ in its code
- Note the use of >= to cover the range correctly

Ex-02 Python SOLUTION

#tutorial-04_ex-02 solution.py

```
score = int(input("Enter score: "))
```

```
if score >= 90:
```

```
    print ("A")
```

```
elif score >= 80:
```

```
    print ("B")
```

```
elif score >= 70:
```

```
    print ("C")
```

```
elif score >= 60:
```

```
    print ("D")
```

```
else:
```

```
    print ("Fail")
```

```
print("Program ended")
```